Machine Learning-Based Classification of Soccer Attacking Phases Using Spatiotemporal Tracking Features

As you wrap up your research and development, it is now time to put your work into words and formulate a final paper to document your progress and findings for the rest of the world to see. This template will serve as your general guide to organizing your thoughts and ideas and transforming your work into a comprehensive scientific paper. Your final paper will be composed of 8 main sections: **Abstract, Introduction, Background, Dataset, Methodology/Models, Results and Discussion, Conclusion, Acknowledgements,** and **References.** Below, each section and its respective composition is described (if you're looking for a more detailed description, **check out this guide**!).

1. Abstract

[< 250 words] The abstract should summarize the key aspects of your research in one paragraph, including motivation, methodology, and the most significant findings. Overall, your abstract is meant to be a concise summary of your research process, from the research question to the results. Remember, non-technical minded people might be reading the abstract, and they should be able to largely understand what's going on! *After filling out the questions below, combine all of your answers to compose your abstract!*

Answer the following questions in 1 or 2 sentences:

1. **What is your research problem or question? What are you trying to accomplish?**

   Can player and ball tracking data be used to classify a soccer team's attacking tactic? I am trying to classify each attacking phase as build-up, creation, direct attack, fast transition, or finishing.

2. **What is the background / context of the problem or question? Why is it important?**

   Coaches and analysts normally identify tactics by manually watching matches. This takes time and can be subjective. A machine-learning model could help organize match footage and find tactical patterns more quickly and consistently.

3. **What was the overall approach used to answer the question?**

I processed tracking data from 10 SkillCorner matches so that each row represented one attacking phase. I created 18 features describing ball movement, team shape, player spacing, pressure, and time. I then trained and compared several classification models using separate training and testing.

4. **What were the significant results? Numbers should be understandable without reading the paper, so expressing differences as fractions or percentages may help the reader.**

Enhanced XGBoost correctly classified about 92 out of every 100 test phases. It reached 91.9% accuracy, a macro F1 of .864, and a macro AUC of 0.990. Enhanced Random Forest reached 91.2% accuracy and a slightly higher macro F1 of 0.873

5. **What are the major conclusions in relation to the problem or question?**

   Tracking data can successfully identify attacking soccer tactics. Ball starting position, forward progress, phase length, and team shape were especially useful. Fast transition was the hardest tactic because the dataset contained fewer examples of it.

6. ***[Your Abstract Section Here]***

This study investigates whether player and ball tracking data can be used to automatically classify attacking phases in soccer. Using SkillCorner Open Data from 10 Australian A-League matches, we processed tracking information into 2,955 attacking phases, each represented by 18 numerical features describing ball movement, team shape, player spacing, defensive pressure, and temporal characteristics. We evaluated multiple supervised machine learning algorithms to classify five attacking phases: build-up, creation, direct attack, fast transition, and finishing. Enhanced XGBoost achieved the highest test accuracy of 91.9%, with a macro F1 score of 0.864, while Enhanced Random Forest achieved the highest macro F1 score of 0.873. Ball starting position, forward progress, and phase duration-related measurements were particularly informative. Fast transitions were the most difficult to classify, likely due to class imbalance and similarities with other attacking phases. These findings demonstrate the potential of interpretable, feature-based machine learning methods to support soccer tactical analysis, although additional evaluation across leagues and matches is necessary to assess generalizability.

2. Introduction

[~ 1 pages long] Clearly discuss your research question, its significance, and any relevant background information needed for the reader to fully understand the problem. Furthermore, highlight the type of problem you are working with (e.g. supervised/unsupervised & regression/classification), the nature of the data you are working with (e.g. numerical/categorical, language data, vision data), and the output of your project (e.g. labels, quantities, etc.).

Soccer tactics are usually identified by coaches or analysts watching match footage. They look at where the players are standing, how quickly the ball moves, and how the team reacts after gaining possession. Reviewing this manually can take a long time, especially when an analyst needs to study several matches. The labels can also be subjective because two people may describe the same attack differently.

We wanted to find out whether tracking data could be used to classify a team's attacking tactic automatically. Our research question was: can player spacing, team shape, ball position, and movement during a possession phase be used to identify the type of attack?

We focused on five attacking tactics. Build-up describes a controlled attack that normally begins farther from the opponent's goal. Creation describes a phase in which the team is developing an attacking opportunity. A direct attack moves the ball forward more quickly. Fast transition happens when a team attacks soon after gaining possession. Finishing describes the final part of an attack near the opponent's goal.

Tactical phase identification was formulated as a supervised multiclass classification problem, where each attacking phase is represented by a numerical feature vector x and assigned a tactical label y from five categories: build-up, creation, direct attack, fast transition, and finishing. The model learns relationships between spatial and temporal features and their corresponding labels to predict the tactical classification of previously unseen attacking phases.

Most of the data used in the model was numerical. It included ball coordinates, distances, time measurements, team width, team depth, player spacing, and the number of defenders near the ball. The tactic label was categorical because it represented one of the five tactic names.

The purpose of the model is not to replace coaches or analysts. Instead, it could help organize match footage and make examples of certain attacks easier to find. This project also allowed us to test whether measurable player and ball movements matched the tactical labels provided by SkillCorner.

The original labels were supplied by SkillCorner through its Phases of Play framework rather than being manually created for this project. SkillCorner uses its own AI-based system to generate these classifications. Therefore, our model acts as a surrogate classifier that attempts to reproduce SkillCorner's labels using a smaller set of tracking features.

3. Background

[~ 0.5 pages long] Describe 1-3 other articles/approaches that have been used to solve your research problem. Briefly highlight pros, cons, or unique contributions of other approaches and how they may relate to your research approach. *Consolidate the work you already completed in your Miniature Literature Review to write this section!*

Tracking data records the locations of players and the ball throughout a match. This gives more information about team shape than event data, which mainly records actions such as passes, shots, fouls, and goals. SkillCorner creates its tracking data from broadcast footage using computer vision [1].

Previous research has shown that artificial intelligence can be used to study team movement and tactical behavior in soccer. A systematic review found that researchers have used machine learning to study positioning, collective movement, and playing styles [6]. This supports our use of measurements such as width, depth, spacing, and ball progress.

Li and Link used a temporal graph model to identify phases of possession [3]. Their model represented players as part of a changing network and studied how their relationships developed over time. This allowed the model to keep information about the order of player movements. One advantage of this approach is that soccer tactics happen over time rather than in one still image. However, the method is more complicated and requires more data and computing power.

Other research has focused specifically on detecting counterattacks [4]. This work showed that features related to speed, direction, and player positioning can help separate counterattacks from other possessions. Research on overlapping tactical actions also showed that one part of a possession may contain more than one tactical idea [5]. For example, a fast transition can later turn into a finishing phase.

This study uses a feature-based classification approach instead of processing entire tracking sequences through complex temporal models. Each attacking phase is summarized using numerical measurements of player positioning, ball movement, and team structure. This reduces the amount of information the models need to process and makes it easier to identify which features contribute most to their predictions. However, averaging measurements across a phase removes information about the exact order of player and ball movements, which may be important when distinguishing rapidly changing tactics.

4. Dataset

[~ 1 page long] Describe the dataset that you used for your project. State the type of data you are working with (numerical, language, vision), the number of samples you have, and how you split the data across training and testing datasets. Explain any data preprocessing procedure you carried out and show any visualizations of your dataset or features (if applicable). Be sure to fully describe the features of your dataset and explain the significance of the features with respect to your research problem. *Consolidate the work you already completed in your Dataset Selection to write this section!*

We used SkillCorner Open Data from 10 Australian A-League matches [1]. The original files included player and ball tracking coordinates, match information, player information, team information, possession information, and phase labels.

The processed dataset contained 2,955 attacking phases and 21 total columns. Eighteen numerical columns were used as model inputs. The tactic_label column contained the correct tactic that the model tried to predict. The other two columns, match_id and phase_index, identified the match and attacking phase and were not used as model inputs.

The dataset contained 1,226 creation phases, 662 finishing phases, 598 build-up phases, 337 direct attacks, and 132 fast transitions. Creation was the largest class, while fast transition was much smaller than the others.

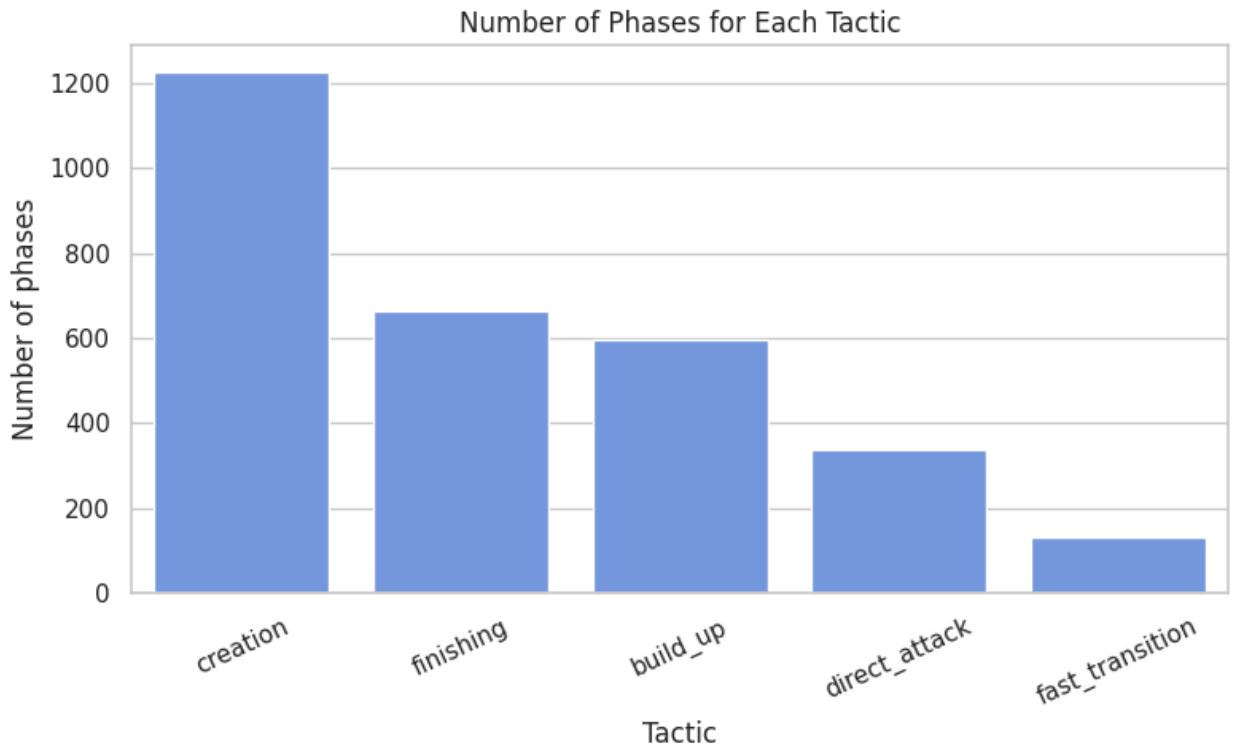


Figure 1. Distribution of attacking-tactic labels in the processed dataset. Creation was the most common label, while fast transition had the fewest examples.

The features described different parts of an attack. Ball features included starting position, forward progress, wide share, and progress speed. Time features included match minute, period, duration, and sampled frames. Team-shape features included attacking and defending width, depth, and spacing. The data also measured nearby defenders, attackers in wide areas, and attackers ahead of the ball.

ball_x_start	ball_progress_m	attack_width_mean	attack_depth_mean	duration	progress_speed_mps	tactic_label
-2.580000	-10.860000	50.140000	48.940000	6.100000	-1.780000	creation
-8.410000	21.190000	55.500000	59.900000	3.500000	6.050000	direct_attack
-23.970000	2.420000	48.790000	49.450000	4.300000	0.560000	build_up
-11.360000	39.600000	47.680000	55.190000	5.100000	7.760000	creation
35.270000	8.950000	37.680000	71.480000	6.700000	1.340000	finishing

Table 1. Sample rows from the processed dataset. Each row represents one attacking phase.

The data was divided by match instead of randomly selecting individual rows. Eight matches were used for training, producing 2,388 phases. Two matches were used for testing, producing 567 phases. Keeping entire matches separate created a more realistic test because the models made predictions on matches they had not seen during training.

Two feature sets were tested. The original feature set used 14 inputs: sampled frames, ball wide share, ball starting position, forward ball progress, attacking width, attacking depth, attacking spacing, defensive width, defensive depth, defensive spacing, defenders within 5 meters, defenders within 10 meters, attackers in wide zones, and attackers ahead of the ball. The enhanced feature set added match minute, period, phase duration, and ball-progress speed, increasing the total to 18 inputs.

5. Methodology/Models

[~ 2 pages long] Describe the methods you used to solve your research problem. State the machine learning algorithms you used and briefly explain how each one works. Describe how you carried out your model learning procedure (e.g. split the dataset into train/test, removed some of the features, input the data into an sklearn model, etc.).

We began with a dummy baseline. This model mostly predicted the most common class. It did not learn tactical patterns, but it gave us a minimum score that the other models needed to beat.

We then tested Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, MLP Classifier, SVC, and XGBoost.

Logistic Regression was used as a simpler comparison model. It creates mathematical boundaries between the classes. Since it mainly learns linear relationships, it may miss more complicated patterns in player movement.

A Decision Tree separates the data through a series of decisions based on the feature values. Decision trees are easy to understand, but one tree can fit the training data too closely. Random Forest reduces this problem by combining many decision trees.

K-Nearest Neighbors compares a new phase with similar phases from the training data. SVC creates boundaries between classes. MLP Classifier is a small neural network that learns relationships between the inputs and labels.

XGBoost builds decision trees one at a time. Each new tree focuses on correcting mistakes made by the earlier trees. This makes it useful for learning complicated relationships between ball movement, player positioning, and team shape.

All models used the same match-based training and testing sets. StandardScaler was used with Logistic Regression, KNN, MLP, and SVC because these models can be affected by the numerical size of each feature. Tree-based models did not require scaling.

Two feature configurations were evaluated during model development. The original configuration used 14 numerical features describing ball movement, player positioning, and team shape. The enhanced configuration included 18 features to provide a more detailed representation of each attacking phase. Both configurations were evaluated using the same match-based training and testing split, allowing their classification performance to be compared. Feature importance analysis and feature-removal experiments were also conducted to determine which measurements contributed most to model performance.

The classes were uneven, so class or sample weights were used when the model supported them. This gave more importance to smaller classes such as fast transition.

We used accuracy and macro F1 to compare the models. Accuracy measures the percentage of all predictions that were correct. Macro F1 calculates an F1 score for each tactic and then averages the five scores equally. This prevented the large creation class from controlling the entire evaluation.

We also calculated precision and recall. Precision measures how often a predicted tactic was correct. Recall measures how many real examples of a tactic were identified.

A confusion matrix showed which tactics the model confused. ROC curves and AUC measured how well the model separated each tactic using its prediction probabilities.

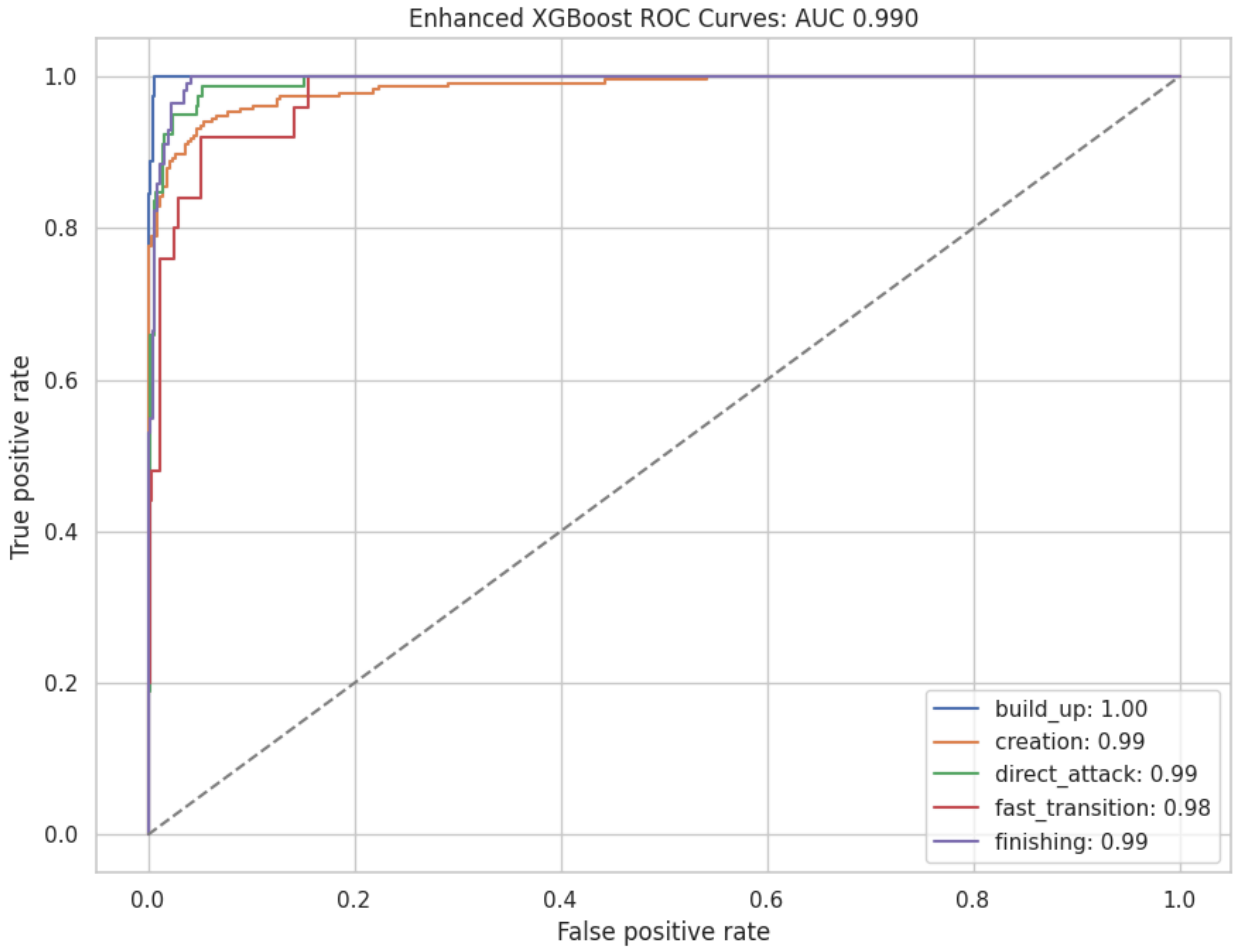


Figure 4. ROC curves for the five attacking tactics. The macro AUC of 0.990 shows that the model separated the classes well.

We tested feature importance using XGBoost importance and permutation importance. Permutation importance shuffled one feature at a time and measured how much the score changed.

We also removed groups of two, three, and five weaker features. The model and testing matches stayed the same, so only the selected features changed.

GroupKFold tested the model on five different groups of matches. This helped us check whether the final result came from one unusually easy testing split.

Finally, we tuned XGBoost using Randomized Search. We tested 25 combinations of tree count, tree depth, learning rate, sampling, minimum child weight, and regularization. Each combination was tested across five folds.

6. Results and Discussion

[~ 2 pages long] Highlight your research findings. Describe the outcomes and metrics of your developed models. Include all relevant model metrics and make sure to highlight any hyperparameters you selected or modified for your models. Include visualizations, tables, and figures to depict your results. Clearly describe all figures and their significance. Furthermore, highlight any errors your method makes and discuss why you think your method may perform poorly in some cases. *Consolidate your Model Metrics and Hyperparameters Sheets to write the latter part of this section!*

Random Forest and XGBoost produced the strongest results. Enhanced XGBoost reached 91.9% accuracy and a macro F1 of 0.864. Original XGBoost reached 91.4% accuracy and a macro F1 of 0.870.

Enhanced Random Forest reached 91.2% accuracy and a macro F1 of 0.873. Original Random Forest reached the same accuracy but had a macro F1 of 0.863.

SVC reached 87.9% accuracy and a macro F1 of 0.825. MLP reached 85.8% accuracy and a macro F1 of 0.790. KNN reached 75.6% accuracy and a macro F1 of 0.649.

Enhanced XGBoost had the highest overall accuracy. Enhanced Random Forest had the highest macro F1, meaning it was slightly more balanced across the five tactics.

Model Performance Comparison

Model	Accuracy	Macro F1
Enhanced XGBoost	0.919	0.864
Original XGBoost	0.914	0.870
Original Random Forest	0.912	0.863
Enhanced Random Forest	0.912	0.873
SVC	0.879	0.825
Logistic Regression	0.863	0.811
MLP	0.858	0.790
Decision Tree	0.814	0.761
KNN	0.756	0.649
Dummy Baseline	0.409	0.116

Table 2. Accuracy and macro F1 scores for all classification models tested.

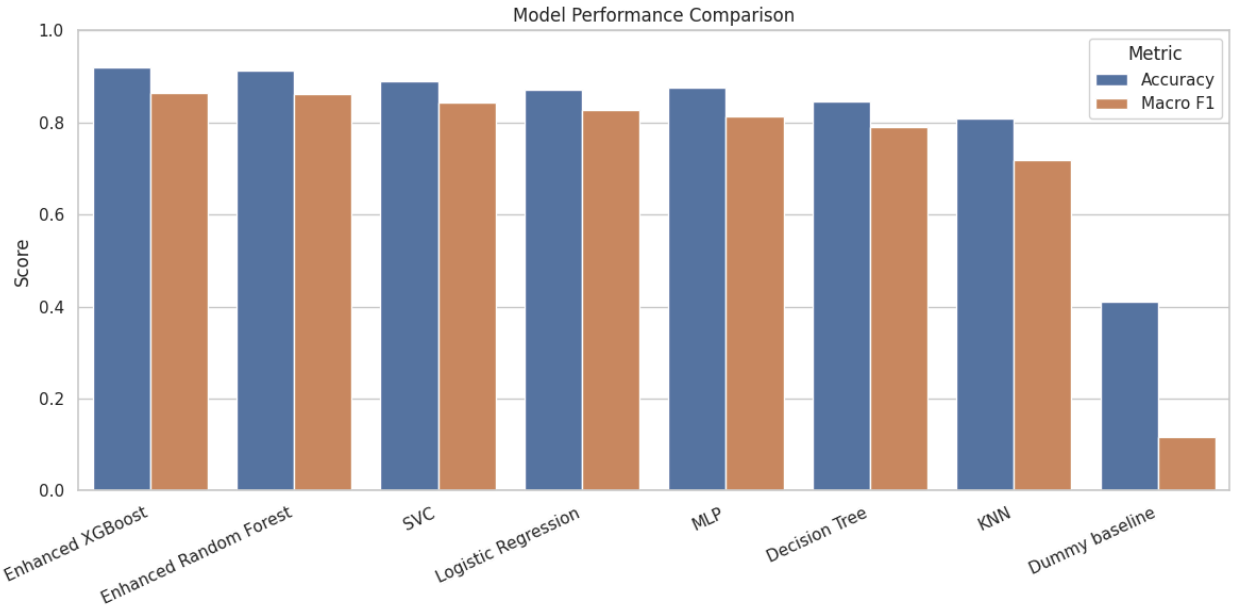


Figure 2. Accuracy and macro F1 scores for the tested models. Random Forest and XGBoost produced the strongest results.

The Enhanced XGBoost model had an F1 score of 0.98 for build-up, 0.92 for creation, 0.91 for direct attack, 0.59 for fast transition, and 0.91 for finishing.

The model correctly classified all 117 build-up phases in the testing data. Creation, direct attack, and finishing also had strong results. Fast transition had the lowest performance. The model correctly identified 13 of the 25 fast-transition phases.

Fast transitions were most frequently misclassified as creation or finishing phases. These categories can share similar spatial and movement characteristics. For example, a fast transition may rapidly advance the ball into the attacking third, producing ball positions and forward progression measurements similar to a creation phase. Similarly, a transition that reaches the opponent's penalty area may resemble a finishing phase. Because the model summarizes each phase using aggregated numerical features, it may not fully capture the sequence of events distinguishing a rapid transition from an established attack. Additionally, fast transition was the least represented class, containing only 132 examples, which may have limited the model's ability to learn its distinguishing characteristics.

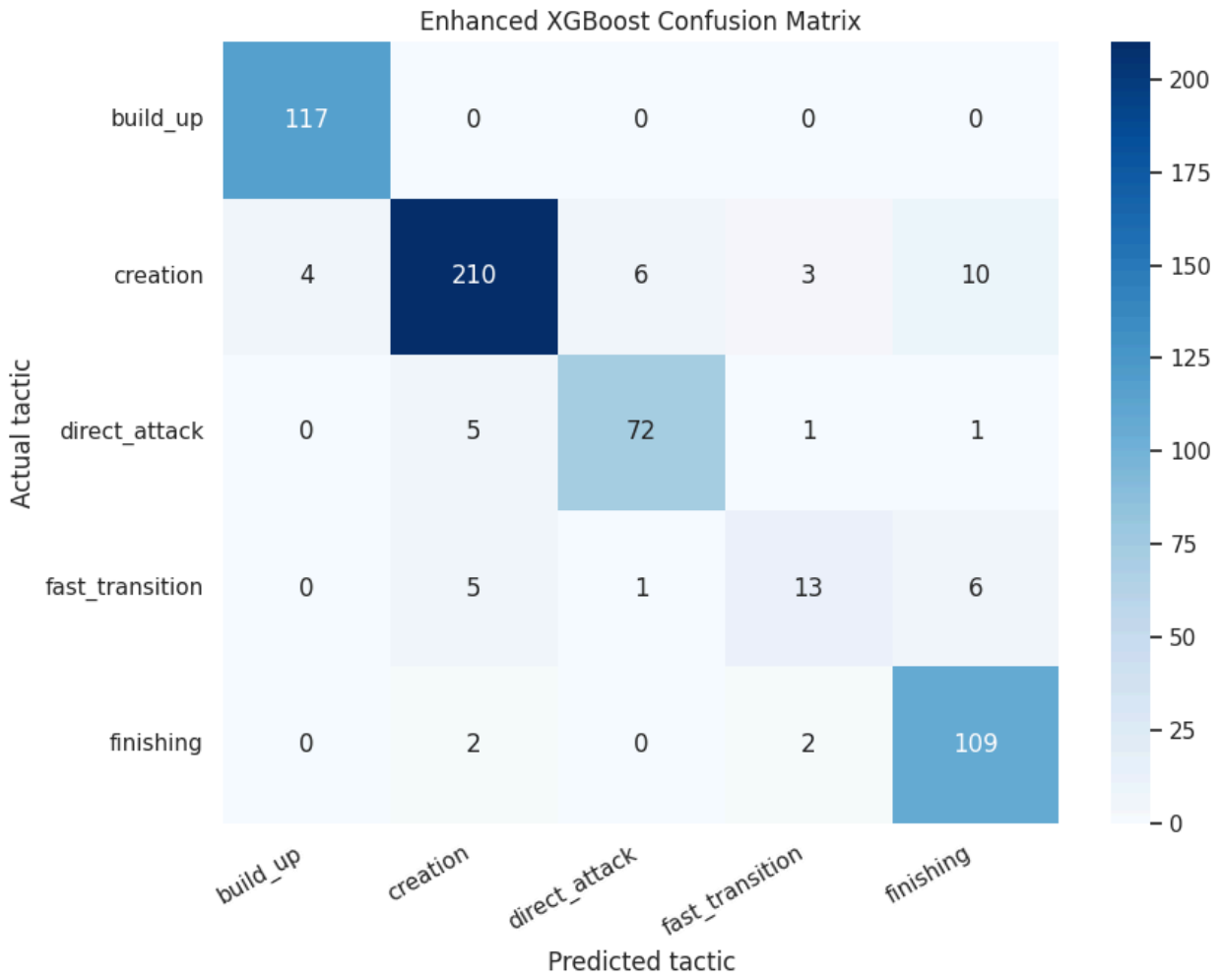


Figure 3. Enhanced XGBoost confusion matrix. Most predictions were correct, but fast transition had the most errors relative to its class size.

The macro AUC was 0.990. An AUC close to 1.00 means that the model separated the classes well using its prediction probabilities.

Ball starting position was the most important feature in Enhanced XGBoost, with an importance score of 0.1961. Other important features included sampled frames at 0.1187, forward ball progress at 0.1124, and attackers ahead of the ball at 0.1018.

Attacking depth, attacking width, progress speed, and defending depth also contributed to the model. Match minute and period had lower importance than most of the spatial features.

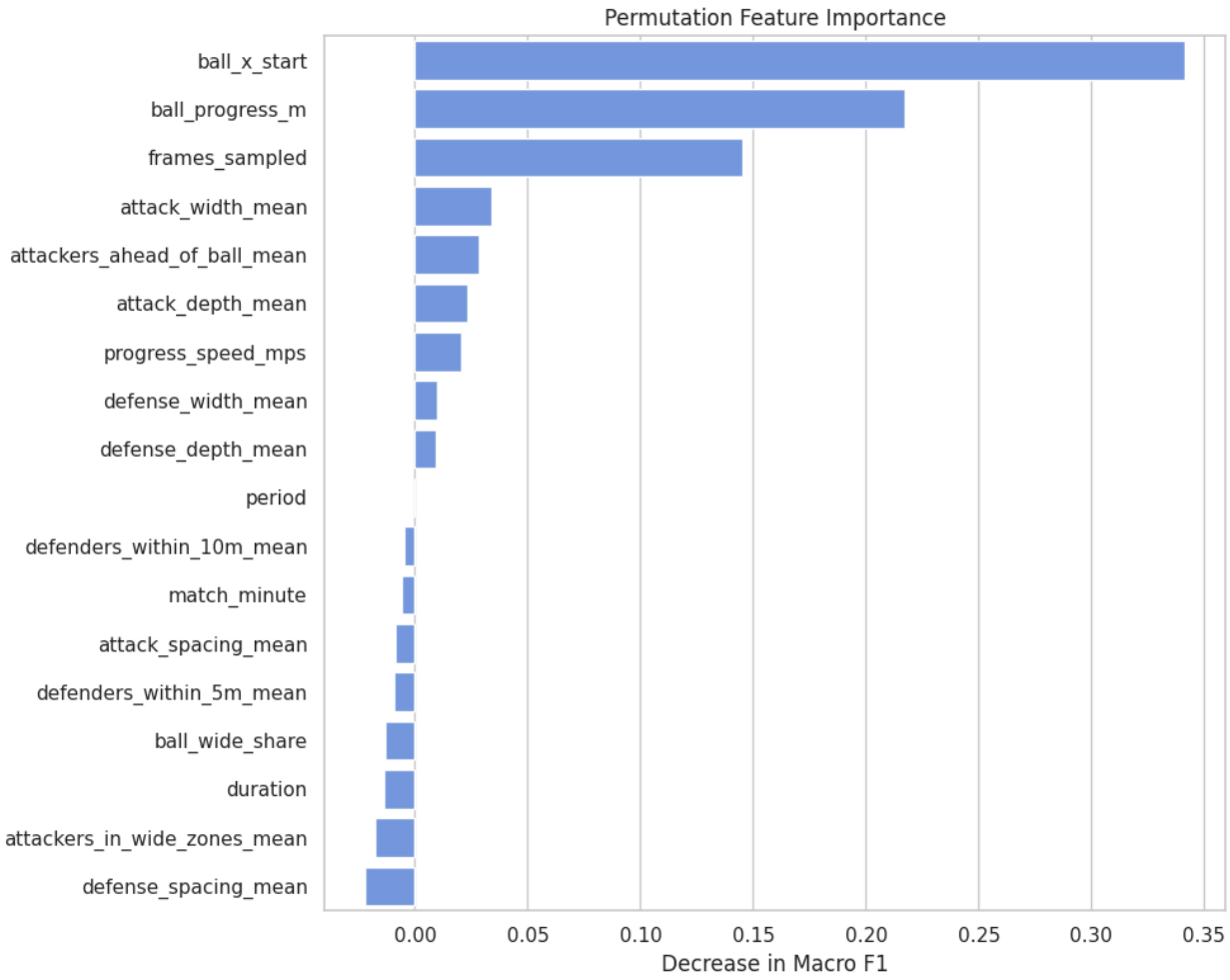


Figure 5. Permutation feature importance. Ball starting position and forward ball progress had the largest effects on model performance.

These results make sense based on how attacks happen. Build-up phases often begin farther from the opponent's goal and move forward more slowly. Direct attacks and fast transitions usually cover more distance in less time. Finishing phases normally happen in advanced areas.

We also tested removing weaker features. Using all 14 original features produced 91.2% accuracy and a macro F1 of 0.863. Removing two produced 91.0% accuracy and a macro F1 of 0.858. Removing three produced 91.0% accuracy and a macro F1 of 0.861. Removing five produced 91.2% accuracy and a macro F1 of 0.867.

Removing five features slightly improved macro F1, but the change was small. The results showed that several weak features could be removed without seriously damaging the model. However, they did not prove that the smaller feature set was clearly better.

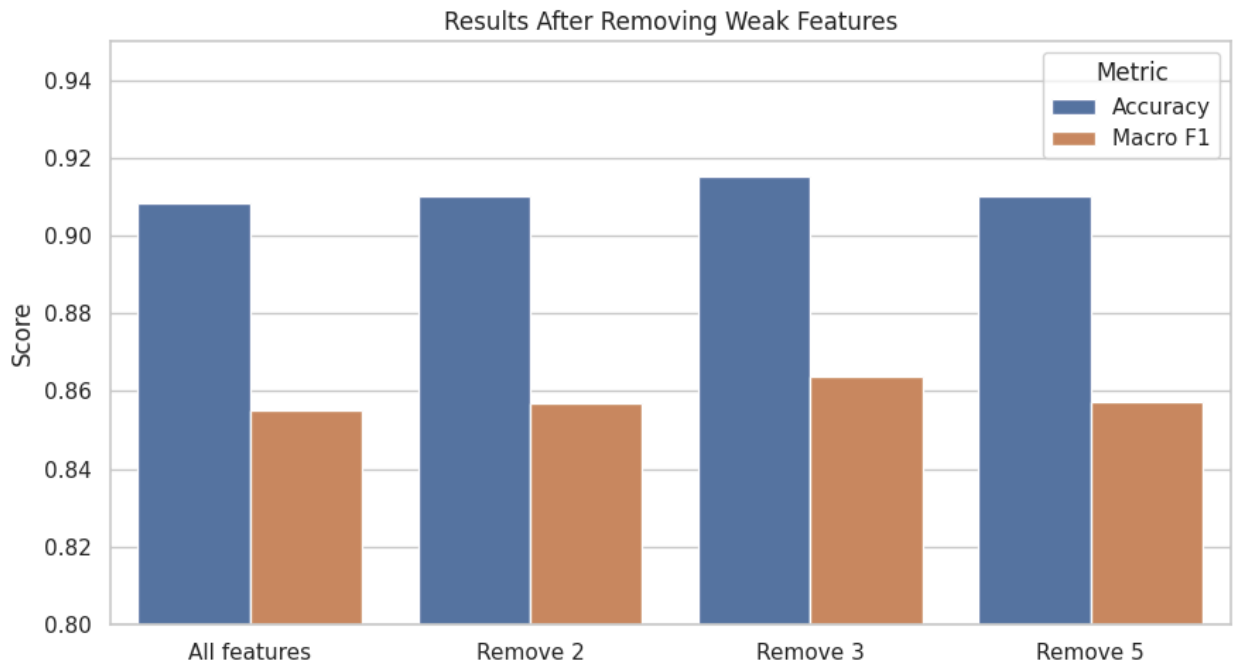


Figure 6. Performance after removing groups of weaker features. Removing five features caused only a small change in macro F1.

GroupKFold produced an average accuracy of around 91% and an average macro F1 of around 0.85. The scores changed between folds, showing that some matches were more difficult than others.

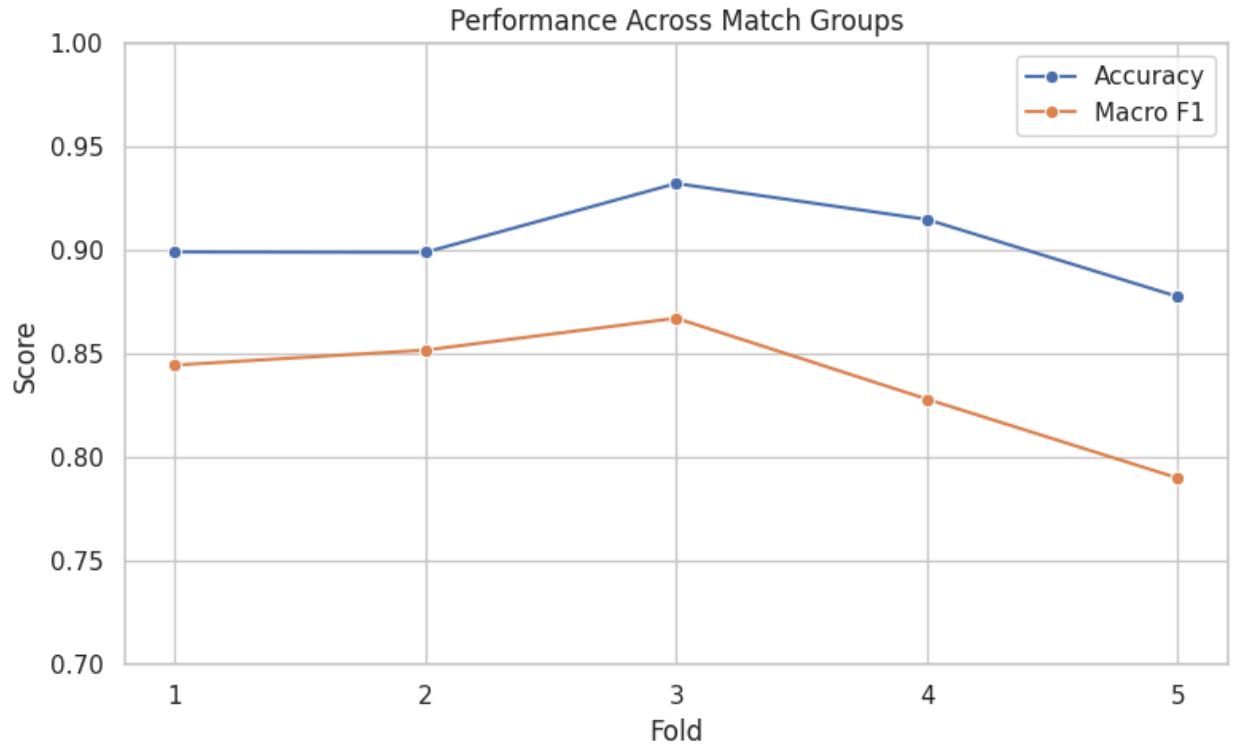


Figure 7. Accuracy and macro F1 across five match-based folds. Performance changed depending on which matches were held out.

Randomized Search produced a best validation macro F1 of 0.884. The best settings were 400 trees, a maximum depth of 7, a learning rate of 0.03, a subsample of 0.8, a column sample of 0.8, a minimum child weight of 5, and regularization of 1.

The results show that tracking features can classify attacking tactics with strong overall accuracy. However, the lower fast-transition score also shows that accuracy does not explain the complete performance of the model.

The dataset only contained 10 matches from one league. Teams in other leagues may use different formations, speeds, and playing styles. The model would need to be tested on more competitions before we could say that it works generally.

Another limitation is that each phase was summarized using averages. This made the model easier to train, but it removed some information about the order of movement during the phase.

7. Conclusion

[~ 0.5 pages long] Summarize your research paper clearly and concisely. Highlight key points regarding your goals, significance, methodology, and results. Describe why you think your model performed as well/poorly as it did. Finally, explain what next steps you would take with your research (e.g. try out different models, obtain more data, preprocess data differently, etc.).

We tested whether player and ball tracking data could classify five attacking tactics in soccer. The dataset contained 2,955 phases from 10 SkillCorner matches, and each phase was described using 18 numerical features.

Random Forest and XGBoost were the strongest models. Enhanced XGBoost reached the highest accuracy at 91.9%. Enhanced Random Forest reached 91.2% accuracy and the highest macro F1 at 0.873.

Ball starting position and forward progress were two of the most important features. The models also used phase length, team width and depth, and the number of attackers ahead of the ball.

The models performed well because the five tactics had noticeable differences in ball location, movement, and team shape. Fast transition was harder because it had fewer examples and shared features with other attacking phases.

The next step would be to collect more matches from different teams and leagues. We would also review more individual correct and incorrect predictions to understand why the models made certain mistakes.

Future features could include the score difference, the previous attacking tactic, and how tactics change during the match. A sequence model could also keep the full order of player and ball movements instead of using only averages from the phase.

8. Acknowledgements

[2-3 sentences max] In this section, you should briefly thank or acknowledge any individuals or institutions that provided significant help and/or advice on your research, analysis, and report writing

I would like to thank my mentor, Omar Ramos Escoto, for guiding me through the research process, helping me improve my models, and providing feedback throughout the project. I also thank Inspirit AI for supporting this research and SkillCorner for making its tracking data publicly available.

9. References

[No page limit] List all the articles and other sources that you used to compile this final paper. Citing your sources and keeping them carefully organized will greatly help in furthering your own research. There are many online citation resources that you can use. Additionally, there are many reference managers that simplify this process. Once you've compiled all your resources, add them here and add any relevant in-text citations.

[1] SkillCorner. "SkillCorner Open Data."
https://github.com/SkillCorner/opendata

[2] SkillCorner. "Phases of Play CSV Specifications."
https://26560301.fs1.hubspotusercontent-eu1.net/hubfs/26560301/Guides/Phases%20of%20Play/20250216%20-%20Phases%20of%20Play%20CSV%20Specifications.pdf

[3] Y. Li and D. Link. "Intention Driven Identification of In-Possession Match Phases in Association Football Through Temporal Graph Learning."
https://arxiv.org/abs/2606.09289

[4] "An AI Framework for Counterattack Detection and Decision-Making Evaluation in Football."
https://link.springer.com/article/10.1186/s40537-025-01128-3

[5] "Estimation of Overlapped Tactical Actions from Soccer Tracking Data."
https://www.scitepress.org/Papers/2024/130663/130663.pdf

[6] "Mapping Football Tactical Behavior and Collective Dynamics with Artificial Intelligence: A Systematic Review."
https://pmc.ncbi.nlm.nih.gov/articles/PMC12163489/